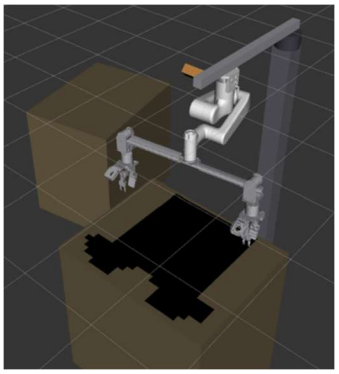
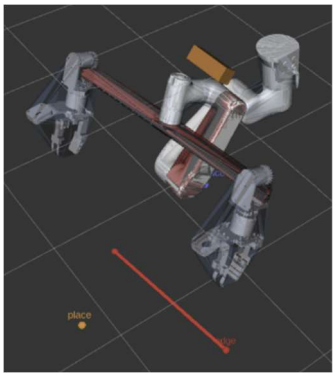
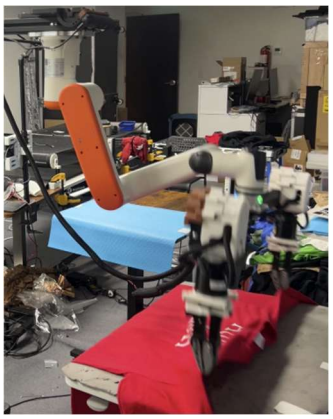
ANA DIMITRIEVSKA – PORTFOLIO

Sim-To-Real Pick and Place – Coop



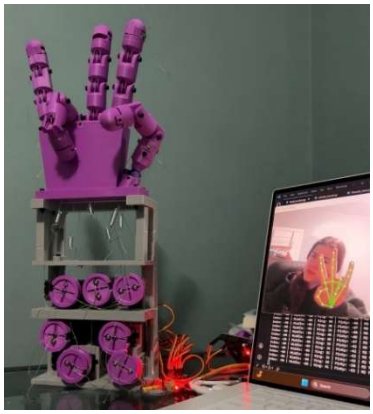
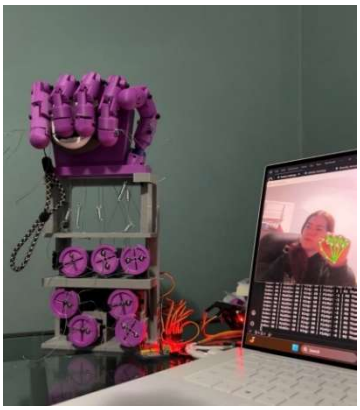
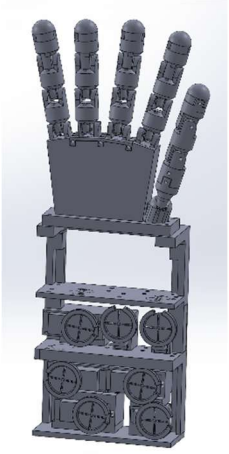
Objective: Develop a **sim-to-real** robotic **pick-and-place** system to automate T-shirt handling using a **RealMan** robotic arm and custom-assembled hardware.

Process: **Assembled** and **wired** the robotic system, including an external emergency stop (E-stop) for safety. Simulated motions in **ROS 2**/**Gazebo**, then connected the physical arm through its **Python API** and **integrated** in-house software for execution. Implemented **collision-aware** path planning and tested potential collisions in simulation before running the robot.

Challenges: Troubleshot **CAN** bus communication issues and **hardware integration**, while debugging discrepancies between simulated and physical robot behaviour.

Results: Successfully deployed the sim-to-real pick-and-place workflow on the physical robot, achieving a **6-second pick cycle** with collision-aware motion planning.

Robot Hand – Personal Project


Objective: Design a robotic hand with human-inspired **kinematics** featuring **1 DOF** per finger and **3 DOFs** for the thumb. An **Arduino Uno** is used for logic and MG995s motors for actuation.

Process: Refined the mechanical design to reduce torque demands, added extension springs, and developed a **Python**/**MediaPipe vision** pipeline using **OpenCV** to track hand motion.

Challenges: Reducing torque requirements for the motors, separating thumb motion into three separate axes, and assembling and manufacturing the full **3D**-printed system.

Results: Developed a functional robotic hand capable of replicating human hand movements and gripping objects in multiple orientations.

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Robot Arm – Personal Project

Objective: Design a robot arm that uses a **vision pipeline** to find the **pose** (3D position and orientation) of an object, **pick** it, and find the pose of its placement location, then **place** the object.



Generated Prototype for Visualization

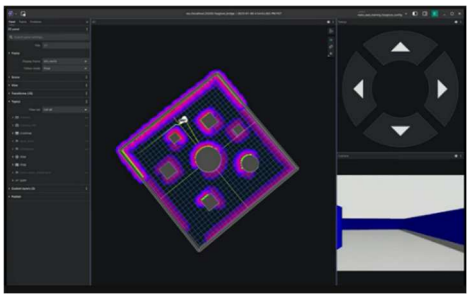
Features:

- **5 DOF** Robot Arm (Rotational Base, 3 Actuated Links, Gripper)
- **Arduino/RaspberryPi**, MG966R Motors
- Monochrome camera for image capture
- **OpenCV** for image processing, **Aruco** marker detection for locating cubes, **PnP solver** to find cube pose
- **ChArUco** Board for Camera Calibration
- **ROS2** and **MoveIt2** used for robot control, **inverse kinematics**, and **path planning**

Process: This project is currently in progress.

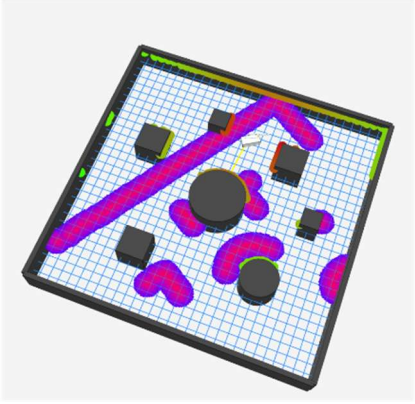
Robot Navigation – Waterloo Watanomous

Objective: Program a simulated differential-drive robot in **ROS2** to autonomously navigate between waypoints while avoiding static obstacles using camera and **LiDAR** data in a **Foxglove** simulation environment.



Process: Researched navigation and obstacle-avoidance methods, integrated sensor data, and refined the system through iterative testing and debugging.

Challenges: Implementing **buffering** and **filtering** systems to improve sensor accuracy and reduce the effects of noise and environmental variability.



Results: Developed a functional autonomous navigation system capable of waypoint navigation and obstacle avoidance in simulation.

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Robot Control Software – Interactive Robotics Lab

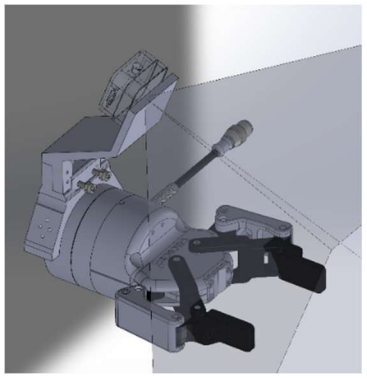
Objective: Develop **ROS2** control software for a **7-DOF PAL Tiago-Pro robotic arm** to perform simulated pick-and-place operations in a **Foxglove** environment.

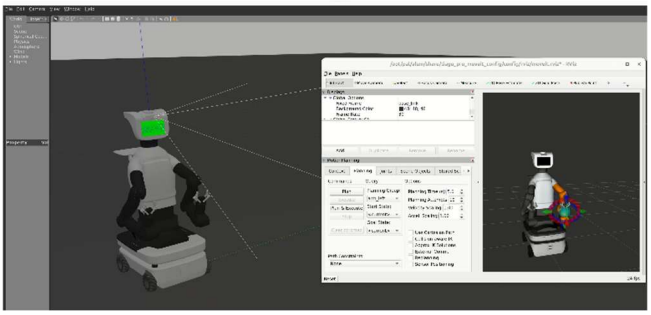
Process: Developed and tested robot motion control, **trajectory planning**, and simulated sensor integration while refining arm movement through iterative testing and debugging.

Challenges: Integrating multiple ROS2 systems, and developing reliable **motion planning** for coordinated arm movement, working with **Docker** and dual booting into **Linux**.

Results: Created a functional simulated pick-and-place workflow with coordinated robotic arm motion in simulation.







Rocket Tower Mount – Waterloo Rocketry

Objective: Design and **machine** a rocket tower mount to support the rocket prior to launch.

Process: Created designs from field measurements, modeled components in **SolidWorks**, performed **FEA** analysis, and validated the design through engine and valve testing.

Challenges: Working under tight design and manufacturing deadlines and adapting through multiple redesigns due to limited reference CAD.

Results: Successfully supported two rocket launches to date.



